Video Game Review Platform

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Problem Statement

The biggest problem we will be facing while making this software is finding out how to deal with unbiased reviews regarding games and companies that have a monopoly over the gaming industry. This problem dates back many years and has been a problem for the gaming community as a whole. Many indie game developers get overshadowed by how bigger reviewers only review games made by AAA companies. That big triple-A company would send free copies or give early access to streamers to make a review, and it will always be a very dishonest review since they usually want to stay in good graces of that company. As a consumer, you want to be able to get honest reviews from people who don't get paid to play the game or get free copies of the game. IGN is a very good example of a company that is very unbiased and true to the actual players. One controversial review they had was for the game "Cuphead," which they gave a poor review because they couldn't beat the tutorial. Our solution is to make software more player-oriented than company-oriented reviews. Some pain points of our project would be: how do we get a database that has up-to-date video game metadata?

Target Users and Stakeholders

Our targeted users would be Video Game enjoyers; these would be the main people using our platform, whether it would be writing game reviews, saving their favorite games, or just seeing what game they would want to play next. These Video game enjoyers can be further broken down into casual and enthusiasts. The casual gamers would want a streamlined experience for tracking and reviewing their favorite games. Meanwhile, gaming enthusiasts are the ones who are going to be writing long, in-depth reviews about recent games. Our biggest

Stakeholders would be Game Developers since they would be looking at our website to see how their game is being portrayed by many people. Game Developers can use our website to see what types of games are performing the best and design their upcoming games based on our reviews. Another stakeholder in our app would be advertising since we will be needing to generate ad revenue to pay for hosting and development fees associated with building our app.

Functional Requirements

FR1 - The system shall let a user create an account with a unique username and password and log in to that account.

FR2 - The system shall display a list of all games whose title contains the text entered in the search box.

FR3 - The system shall display a "No matches found" message followed by a list of trending and newly released games when a search returns no results.

FR4 - The system shall require a reviewer to state whether they received the game for free or were paid to review it, and shall display that disclosure on the review.

FR5 - The system shall let a logged-in user cast one upvote or downvote on any review written by another user.

FR6 - The system shall display a game's reviews in descending order of net votes.

FR7 - The system shall let an administrator add a game and edit its details (title, genre, developer, release date).

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Non-Functional Requirements

NFR1 - The system shall return search results within 300 milliseconds for at least 95% of searches under a load of 100 concurrent users.

NFR2 - The system shall display the fallback list within 500 milliseconds of the search request.

NFR3 - The user interface shall be structured such that a user can navigate to any game category within a maximum of 3 clicks from the home screen.

NFR4 - The system shall process and save all user-generated actions that are committed and reflected across all clients within 1.0 seconds without data loss.

NFR5 - The system shall be able to handle a peak load of at least 100 concurrent users without dropping connections.

2.1 User Stories

FR –1

User Story: As a casual gamer, I want to search for a game by its title so that I can quickly find and read its reviews.

Acceptance Criterion: Given a game titled "Cuphead" exists in the database, when I search for "Cuphead", then the system displays "Cuphead" in the search results with a link to its review page.

FR – 5

User Story: As a registered gamer, I want to organize and manage my saved games in custom lists so that I can keep track of the games I have reviewed or want to play.

Acceptance Criterion: Given I am logged into my account, when I add a game to my "Favorites" list, then the game appears in my personal dashboard under the "Favorites" category.

FR – 6

User Story: As a registered gamer, I want to upvote or downvote user reviews so that top-rated helpful reviews appear at the top and poor reviews move to the bottom.

Acceptance Criterion 1: Given a published review written by another user, when I select Upvote, then the review's vote count increases by 1.

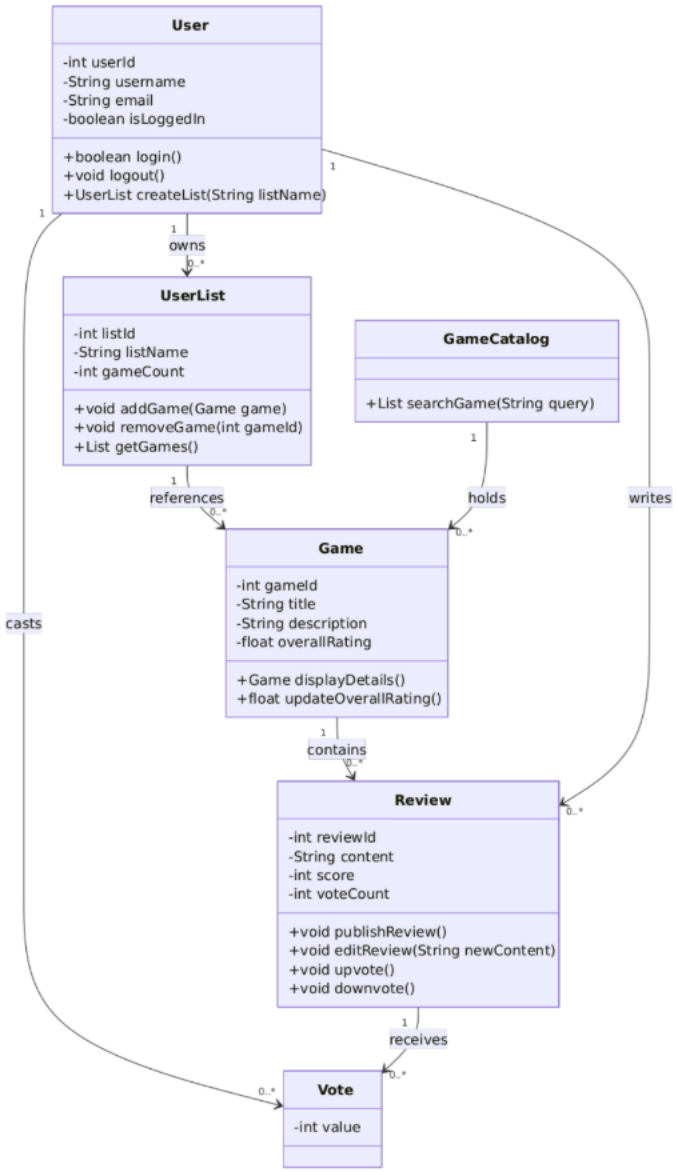
Acceptance Criterion 2: Given a list of published reviews on a game page, when a review's vote count changes, then the list re-sorts to reflect the updated vote total.

NF – 3

User Story: As a video game enjoyer, I want to navigate to any game category from the home screen within 3 clicks so that I can find content quickly without excessive clicking.

Acceptance Criterion: Given I am on the home screen, when I navigate through category links, then I can reach any specific game category page in 3 or fewer clicks.

Class Diagram

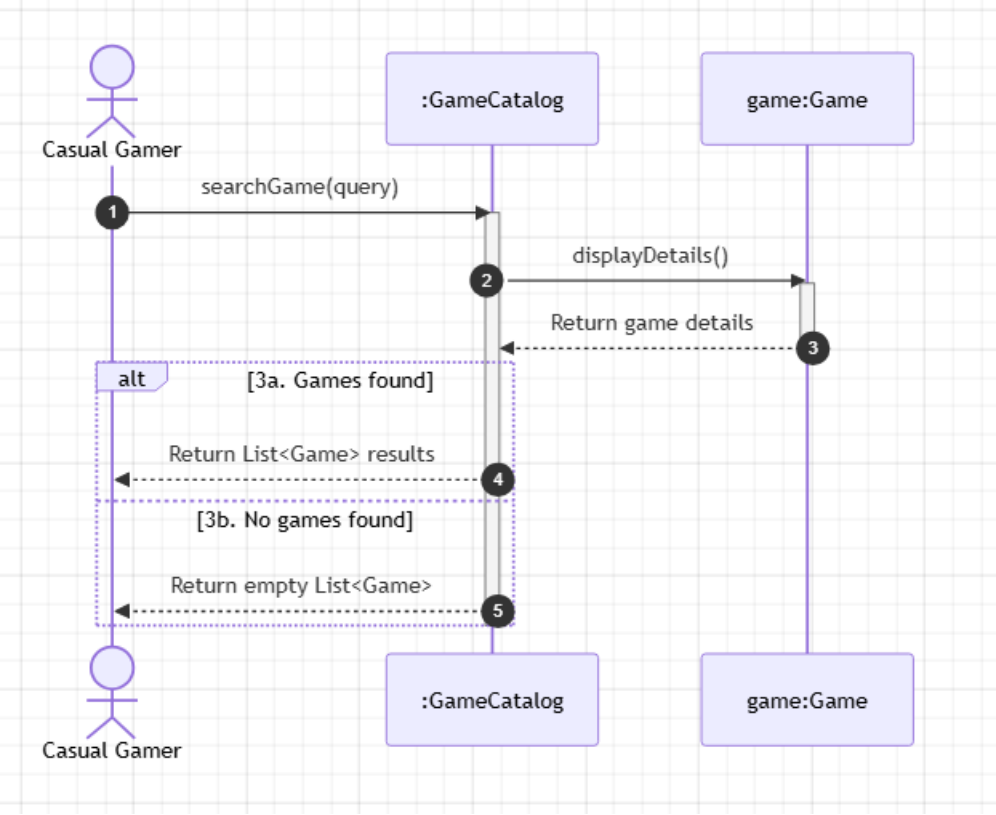


2.2 Table A – Scenarios

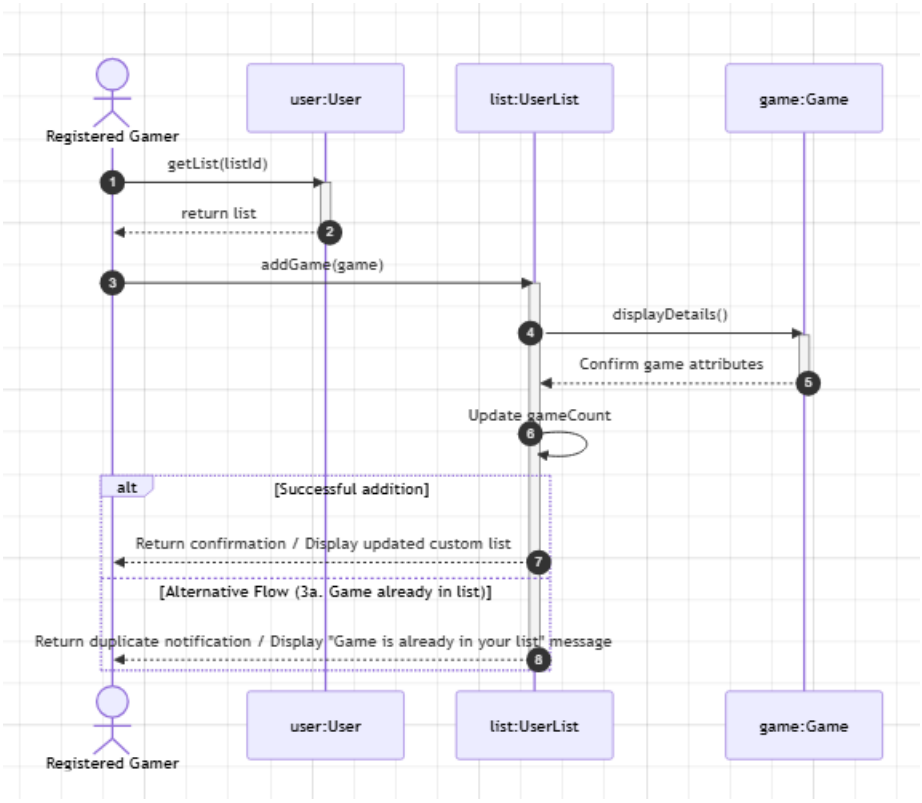
#	User Story	Main Flow	One alternative flow
FR-1	As a casual gamer, I want to search for a game by its title so that I can quickly find and read its reviews.	1. The user types the title of the game they are looking for into the search bar 2. The search interface passes the search text over to the GameCatalog.	3a. If no matching games exist in the database, the catalog returns an empty list, and the system prompts the user with a "No games found" message.

		3. The catalog checks its records, finds the matching games, and pulls their details. 4. The system delivers the search results, complete with direct links to each game's review page.	
FR-5	As a registered gamer, I want to add a game to one of my custom lists so that I can keep track of the games I want to play	1. The user chooses a game and clicks the button to save it to their custom list. 2. The user dashboard sends an add request directly to the target UserList. 3. The list verifies the game's details and updates its internal game count. 4. The system successfully saves the game to the collection and refreshes the user's view.	3a. If the game is already saved in that list, the system blocks the duplicate entry and displays a message letting the user know it is already there.

Diagrams A1 -



Diagrams A2 -


Registered Gamer
user:User
list:UserList
game:Game
getList(listId)
1
return list
2
addGame(game)
3
displayDetails()
4
Confirm game attributes
5
Update gameCount
6
alt
[Successful addition]
Return confirmation / Display updated custom list
7
[Alternative Flow (3a. Game already in list)]
Return duplicate notification / Display "Game is already in your list" message
8
Registered Gamer
user:User
list:UserList
game:Game